

**Tutorial 9**  
**Chapter 10**

**Monday:**

Problem 13  
Problem 60 a

**Tuesday:**

Problem 14 a, b, c  
Problem 61

**Wednesday**

Problem 17 a, b, c  
Problem 60 a

### Problem 13

Initial angular velocity  $\omega_0 = +1.5 \text{ rad/s} = +0.239 \text{ rev/s}$  at  $t = 0$ ,

Constant value of angular acceleration  $\alpha < 0$ , which indicates our choice for positive sense of rotation.

At  $t_1$  its angular displacement (relative to its orientation at  $t = 0$ ) is  $\theta_1 = +20 \text{ rev}$ ,

At  $t_2$  its angular displacement is  $\theta_2 = +40 \text{ rev}$  and its angular velocity is  $\omega_2 = 0$ .

You can use rev or radians in this case.

(a) We obtain  $t_2$  using the equation for angular displacement:

$$\begin{aligned}\theta_2 &= \frac{1}{2}(\omega_0 + \omega_2)t_2 \\ \Rightarrow t_2 &= \frac{2(40 \text{ rev})}{0.239 \text{ rev/s}} = 335 \text{ s}\end{aligned}$$

which we round off to  $t_2 \approx 3.4 \times 10^2 \text{ s}$ .

(b) Any equation in Table 10-1 involving  $\alpha$  can be used to find the angular acceleration; we select the following:

$$\begin{aligned}\theta_2 &= \omega_2 t_2 - \frac{1}{2} \alpha t_2^2 \\ \Rightarrow \alpha &= -\frac{2(40 \text{ rev})}{(335 \text{ s})^2} = -7.12 \times 10^{-4} \text{ rev/s}^2\end{aligned}$$

which converts to  $\alpha = -4.5 \times 10^{-3} \text{ rad/s}^2$  if you used radians in all your variables.

(c) Using  $\theta_1 = \omega_0 t_1 + \frac{1}{2} \alpha t_1^2$  (Eq. 10-13) and the quadratic formula, we have

$$t_1 = \frac{-\omega_0 \pm \sqrt{\omega_0^2 + 2\theta_1\alpha}}{\alpha} = \frac{-(0.239 \text{ rev/s}) \pm \sqrt{(0.239 \text{ rev/s})^2 + 2(20 \text{ rev})(-7.12 \times 10^{-4} \text{ rev/s}^2)}}{-7.12 \times 10^{-4} \text{ rev/s}^2}$$

which yields two positive roots: 98 s and 572 s. Since the question makes sense only if  $t_1 < t_2$  we conclude the correct result is  $t_1 = 98 \text{ s}$ .

You could also use a two step process where you first determine the angular speed after 20 rev, using  $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$ , followed by the calculation of time using

$$\omega = \omega_0 + \alpha t$$

### Problem 14

The wheel starts turning from rest ( $\omega_0 = 0$ ) at  $t = 0$ ,

It accelerates uniformly at  $\alpha > 0$ , which makes our choice for positive sense of rotation.

At  $t_1$  its angular velocity is  $\omega_1 = +10 \text{ rev/s}$ ,

At  $t_2$  its angular velocity is  $\omega_2 = +15 \text{ rev/s}$ .

Between  $t_1$  and  $t_2$  it turns through  $\Delta\theta = 60 \text{ rev}$ , where  $t_2 - t_1 = \Delta t$ .

(a) We find  $\alpha$  using:

$$\begin{aligned}\omega_2^2 &= \omega_1^2 + 2\alpha\Delta\theta \\ \therefore \alpha &= \frac{(15 \text{ rev/s})^2 - (10 \text{ rev/s})^2}{2(60 \text{ rev})} \checkmark \checkmark \\ \alpha &= 1.04 \text{ rev/s}^2\end{aligned}$$

which we round off to  $1.0 \text{ rev/s}^2$ .

(b) We find  $\Delta t$  using:

$$\begin{aligned}\Delta\theta &= \frac{1}{2}(\omega_1 + \omega_2)\Delta t \\ \Rightarrow \Delta t &= \frac{2(60 \text{ rev})}{10 \text{ rev/s} + 15 \text{ rev/s}} \checkmark \checkmark \\ \Delta t &= 4.8 \text{ s}.\end{aligned}$$

(c) We obtain  $t_1$  using:

$$\begin{aligned}\omega_1 &= \omega_0 + \alpha t_1 \\ \Rightarrow t_1 &= \frac{10 \text{ rev/s}}{1.04 \text{ rev/s}^2} \checkmark \checkmark \\ t_1 &= 9.6 \text{ s}.\end{aligned}$$

### Problem 17

The problem has (implicitly) specified the positive sense of rotation. The angular acceleration of magnitude  $0.25 \text{ rad/s}^2$  in the negative direction is assumed to be constant over a large time interval, including negative values (for  $t$ ).

(a) We specify  $\theta_{\max}$  with the condition  $\omega = 0$   $\checkmark$  (this is when the wheel reverses from positive rotation to rotation in the negative direction). We obtain  $\theta_{\max}$  using:

$$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta = \omega_0^2 + 2\alpha(\theta_{\max} - \theta_0)$$

$$\theta_{\max} = -\frac{\omega_0^2}{2\alpha} = -\frac{(4.7 \text{ rad/s})^2}{2(-0.25 \text{ rad/s}^2)} \quad \checkmark \checkmark$$

$$= 44 \text{ rad.}$$

(b) We find values for  $t_1$  when the angular displacement (relative to its orientation at  $t = 0$ ) is  $\theta_1 = 22 \text{ rad}$  (or 22.09 rad if we wish to keep track of accurate values in all intermediate steps and only round off on the final answers). Using the equation for angular displacement and the quadratic formula, we have

$$\theta_1 = \omega_0 t_1 + \frac{1}{2} \alpha t_1^2$$

$$\Rightarrow t_1 = \frac{-\omega_0 \pm \sqrt{\omega_0^2 + 2\theta_1 \alpha}}{\alpha} \quad \checkmark$$

which yields the two roots 5.5 s and 32 s. Thus, the first time the reference line will be at  $\theta_1 = 22 \text{ rad}$  is  $t = 5.5 \text{ s}$ .  $\checkmark$

(c) The second time the reference line will be at  $\theta_1 = 22 \text{ rad}$  is  $t = 32 \text{ s}$ .  $\checkmark$

### Problem 60

(a) We apply the formula for rotational Kinetic energy, and using the moment of inertia for a thin rod and the parallel axis theorem:

$$I = \frac{1}{12} mL^2 + m\left(\frac{L}{2}\right)^2 = mL^2\left(\frac{1}{12} + \frac{1}{4}\right) = \frac{1}{3} mL^2 \quad \checkmark \checkmark$$

$$K = \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{1}{3} mL^2\right) \omega^2 = \frac{1}{6} mL^2 \omega^2 \quad \checkmark \checkmark$$

$$= \frac{1}{6} (0.42 \text{ kg}) ((0.75 \text{ m})^2 (3.5 \text{ rad/s})^2) = 0.48 \text{ J}$$

### Problem 61

The initial angular speed is  $\omega = (280 \text{ rev/min})(2\pi/60) = 29.3 \text{ rad/s}$ .

(a) The rotational inertia is:

$$I = (32 \text{ kg})(1.2 \text{ m})^2 = 46.1 \text{ kg} \cdot \text{m}^2 \quad \checkmark$$

the work done is

$$\begin{aligned}
 W &= \Delta K = 0 - \frac{1}{2} I \omega^2 \\
 &= -\frac{1}{2} (46.1 \text{ kg} \cdot \text{m}^2) (29.3 \text{ rad/s})^2 \quad . \checkmark \checkmark \\
 &= -1.98 \times 10^4 \text{ J}
 \end{aligned}$$

(b) The average power (in absolute value) is therefore

$$P = \frac{|W|}{\Delta t} = \frac{1.98 \times 10^4 \text{ J}}{15.0 \text{ s}} \quad \checkmark$$